

Analysis 2
Serie 8
Lukas Mengis

Jordan-null set.

(a) Let $X \subset \mathbb{R}$ be a Jordan-null set. $\forall \delta > 0, \exists$ a collection of finitely many intervals such that

$$X \subset \bigcup_{j=1}^m I_j \quad \wedge \quad \sum_{j=1}^m |I_j| < \delta$$

Then

$$X \times X \subset \bigcup_{j=1}^m \bigcup_{i=1}^m (I_i \times I_j)$$
$$\sum_{j=1}^m \sum_{i=1}^m |I_j| |I_i| = \left(\sum_{j=1}^m |I_j| \right) \left(\sum_{i=1}^m |I_i| \right) < \delta^2$$

Set $\varepsilon > 0$ and $\delta = \sqrt{\varepsilon}$

$$\sum_{j,i} \text{vol}(I_i \times I_j) < \varepsilon$$

Therefore $X \times X$ can be covered by finitely many squares with volume smaller than ε .

□

(a) Since $X, [0, 1] \subset \mathbb{R}$ Jordan-null set $\exists$ finitely many intervals I_i and I_j covering X and $[0, 1]$

$$X \subset \bigcup_{i=1}^m I_i \quad \wedge \quad \sum_{i=1}^m |I_i| < \varepsilon$$

it follows

$$X \times [0, 1] \subset \bigcup_{i=1}^m (I_i \times [0, 1])$$

Ever $I_i \times [0, 1]$ is a rectangle with volume

$$\text{vol}(I_i \times [0, 1]) = |I_i| \cdot 1 = |I_i|$$

therefore

$$\text{vol}(X \times [0, 1]) \leq \sum_{i=1}^m |I_i| < \varepsilon$$

□

True or False.

1. true
2. true
3. true
4. true
5. false

Fat Boundary.

An open self similar fractal.

Define

$$\begin{aligned} S : \mathbb{R}^n &\rightarrow \mathbb{R}^n \\ x &\mapsto S(x) \end{aligned}$$

Such that $x \cap S(x) = \emptyset$.

Let U_0 be some open subset of $\mathbb{R}^n$.

$$\begin{aligned} U_m &:= \bigcup_{i=0}^{m-1} S_i(U_0) \\ U &= \lim_{m \rightarrow \infty} U_m \end{aligned}$$

Then U is open.

For every $\varepsilon > 0$, there does not exist a finite family of sets $A_1, \dots, A_k$ with $A_j < \varepsilon$ such that

$$\partial U \subset \bigcup_{j=1}^k A_j$$

therefore the ∂U is not a null set.

Multiple Choice.

1. false
2. false
3. false
4. true

Change of variables and Jacobians.

1.
$$x_1 = r \cos(\theta), \quad x_2 = r \sin(\theta), \quad \left| \det \left(\begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{pmatrix} \right) \right| = |r|$$

$$x_1 > 0 \Rightarrow -\frac{\pi}{2} < \theta < \frac{\pi}{2}, \quad x_1^2 + x_2^2 < 4 \Rightarrow r \in (-2, 2), \quad x_2 < x_1 \Rightarrow \left(-3\frac{\pi}{4}\right) < \theta < \frac{\pi}{4}$$

$$x_1^2 \sin(x_2) dx_1 dx_2 = r^3 \cos^2(\theta) \sin(r \sin(\theta)) dr d\theta$$

$$\int_A x_1^2 \sin(x_2) dx_1 dx_2 = \int_{-\frac{\pi}{2}}^{\frac{\pi}{4}} \int_{-2}^2 r^3 \cos^2(\theta) \sin(r \sin(\theta)) dr d\theta$$

2.
$$u = xy \quad v = x^2 \quad \left| \det \left(\begin{pmatrix} y & 2x \\ x & 0 \end{pmatrix} \right) \right| = 2x^2$$

$$1 < u < 2, \quad v < \frac{u}{\sqrt{v}} < 2v \Rightarrow \left(\frac{u}{2}\right)^{\frac{2}{3}} < v < u^{\frac{2}{3}}$$

$$y^2 e^{-xy} dx dy = \frac{1}{2v} \frac{u^2}{v} e^{-u} dv du = \frac{u^2}{2v^2} e^{-u} dv du$$

$$\int_1^2 \int_{\left(\frac{u}{2}\right)^{\frac{2}{3}}}^{u^{\frac{2}{3}}} \frac{u^2}{2v^2} e^{-u} dv du$$

3.

$$u = z, \quad v = z - 2y, \quad w = 3xyz, \quad \left| \det \begin{pmatrix} 0 & 0 & 1 \\ 0 & -2 & 1 \\ 3 & 1 & 1 \end{pmatrix} \right| = 6$$

$$xyz = u \frac{u-v}{2} \frac{w-u-\frac{u-v}{2}}{3} = \frac{u(u-v)(2w-3u-v)}{12}$$

$$\int_C xyz \, dx dy dz = \int_0^1 \int_1^2 \int_{-2}^0 \frac{u(u-v)(2w-3u-v)}{72} du dv dw$$

The Cantor set.

1. We want to define a set F who behaves like a fractal.

$$x \in [0, 1] \text{ and } x = 0.a_1 a_2 \dots \text{ with } a_k \in \{0, 1, \dots, 9\} \quad \forall k$$

$$F = \{x \in [0, 1] \mid a_k \neq 8, \forall k\}$$

This set is a fractal since there is one rule which defines the next decimal place.

After n iterations there are 9^n (with a length of 10^{-n}) possibilities out of 10^n and since

$$\lim_{n \rightarrow \infty} \left(\frac{9}{10} \right)^n = 0$$

This means the set F is Lebesgue null.

2. Define F like in the previous question. Assume we have the list L of all number contained in this set. Claim we always find a number $n \in F$ but $n \notin L$.

This can be done by Cantor's diagonal argument (always 8 numbers to chose from).

3. The same proof as in the first exercise but replace m with ∞ .

4. Define F_n

$$F_n = \bigcup_{k \neq 8} [0, a_1 a_2 \dots a_{n-1} k, 0, a_1 a_2 \dots a_{n-1} (k + 0.\bar{9})]$$

This subset is compact. Every intersection of compact sets is compact.

$$F = \bigcap_{n=1}^{\infty} F_n$$

This set F is compact and does not contain a decimal place which is equal to 8.

□